

Three Mechanisms Integrating Frame: Decomposing B1.12 Three Evolution Mechanisms in Productive Tension by Formalizing the Architectural Framework That Unifies Mutation (B1.13), Directed Selection (B1.14), and Action-Feedback (B1.15) as One Evolution Architecture Covering Undirected Variation, Directed Improvement, and Evidence-Informed Refinement

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 12, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

Abstract

The CKS theory specifies three evolution mechanisms — mutation (instinct evolution), directed selection (DNA evolution), and action-feedback evolution — as operating together in productive tension under unified human governance. This note formalizes the three mechanisms integrating frame: the architectural specification that makes the three mechanisms one evolution architecture rather than three independent processes. The integrating frame is what gives the multi-mechanism evolutionary architecture its coherence. It specifies that the three mechanisms cover distinct and complementary evolutionary dimensions (undirected variation, directed improvement, evidence-informed refinement), operate on architecturally separated layers within CKS, are each governed with mechanism-appropriate shapes, can operate concurrently without mutual interference, apply at every structural level per the recursive-levels commitment, and produce DNA changes subject to A6.02 retroactivity treatment. Without the integrating frame, the three mechanisms would be three uncoordinated processes; the frame is what makes them composable as one architecture. This note opens a five-note B1.12 decomposition (B2.56–B2.60) and is the fifty-sixth note in Phase B2 of the Series B derivation chain from Paper 2.

1. Why the three mechanisms integrating frame requires standalone formalization

Paper 2 (“The Instinct/Reasoning Separation Outside the Model,” Li, April 2026) commits to three evolution mechanisms operating in productive tension under unified human governance. The three mechanisms are identified and individually specified: mutation (instinct evolution) as the mechanism by which the instinct layer evolves through upstream LLM and infrastructure upgrades; directed selection (DNA evolution) as the mechanism by which the DNA layer evolves through

human-governed orchestration substrate changes; and action-feedback evolution as the mechanism by which accumulated operational evidence closes the loop back into governed DNA refinement. These three mechanisms are named, their operation is described, and their governance shapes are differentiated. What Paper 2 does not do — and what the B1.12 decomposition must do — is formalize the integrating frame itself as a standalone architectural specification.

The integrating frame is not one of the three mechanisms. It is the architectural specification that makes the three mechanisms one evolution architecture rather than three coexisting but unrelated processes. A CKS deployment that has all three mechanisms without the integrating frame has three evolution processes that happen to run in the same system. The integrating frame specifies how the three mechanisms relate to each other, how they differ from each other, and under what conditions they can operate together without collapsing into a single mechanism or interfering with each other's operation. This specification is patentable as a distinct architectural commitment separate from the individual mechanism specifications (B1.13, B1.14, B1.15 in Series B numbering), and formalizing it as prior art is the strategic purpose of this note.

The position of B2.56 in Phase B2 reflects this logic. The B1.11 decomposition (B2.51–B2.55) completed the death mechanism treatment. The B1.12 decomposition now begins at B2.56 with the integrating frame — the architectural object that holds the three mechanisms together — before the individual mechanism decompositions proceed. Subsequent B1.12 decomposition notes are: B2.57 (productive tension operational specification), B2.58 (mechanism priority and sequencing), B2.59 (cross-mechanism governance), and B2.60 (three mechanisms verification). After B2.60, Phase B2 continues with the B1.13 mutation decomposition at B2.61–B2.66.

2. The architectural frame precisely stated

The three mechanisms integrating frame specifies six properties that together constitute what it means for three evolution mechanisms to operate as one evolution architecture.

First: The three mechanisms cover distinct and complementary evolutionary dimensions. Mutation (instinct evolution) covers undirected variation. The instinct layer — the LLM and substrate-platform infrastructure — evolves through upstream upgrades that arrive from outside the deployment's governance perimeter. The variation is undirected in the precise sense that the deployment cannot govern what a given LLM upgrade changes; it can only govern how the upgrade is integrated once it arrives. Directed selection (DNA evolution) covers directed improvement. The DNA layer — the stabilized orchestration substrate content — evolves through human deliberate authoring under governance-defined goals. The improvement is directed in the precise sense that humans choose what to improve and set the selection criteria for what changes are accepted. Action-feedback evolution covers evidence-informed refinement. The DNA layer evolves from accumulated operational evidence surfaced through the action layer, with human-mediated proposals and approval governing what changes reach the substrate. The refinement is evidence-informed in the precise sense that proposals are grounded in observed operational patterns rather than in abstract design decisions alone. The three dimensions are complementary: they do not overlap, and no one of them can substitute for either of the others.

Second: All three mechanisms operate within the CKS architecture on architecturally separated layers. Mutation operates on the instinct layer (LLM and substrate-platform infrastructure). Directed selection operates on the DNA layer. Action-feedback operates on the DNA layer through accumulated action evidence. The layer separation is what permits all three mechanisms to operate without structural interference — they do not compete for the same architectural object.

Third: All three mechanisms are governed, with different governance shapes reflecting their different operational character. Mutation is governed through verification gates, routing rules, and pinning of high-stakes decisions to the reasoning layer — governance that responds to undirected variation without pretending to direct it. Directed selection is governed through the standard authority architecture inherited from Paper 1 — governance that specifies who can author what orchestration changes under what conditions. Action-feedback is governed through two-stage human mediation: humans govern the substrates that propose DNA changes from action evidence, and humans approve the changes — governance that prevents the action layer from silently drifting the DNA layer over time. The integration frame specifies that all three mechanisms are governed; the mechanism-appropriate shape of each is the subject of B2.59 (cross-mechanism governance).

Fourth: All three mechanisms can operate concurrently. Mutation does not block directed selection. Action-feedback proposals do not block mutation integration. Directed selection changes do not require action-feedback to pause. The concurrent operation is architecturally available because the mechanisms operate on separated layers (first property) and because governance for each is specified independently (third property). An active deployment can be simultaneously integrating an LLM upgrade, authoring a DNA layer change, and processing an action-feedback proposal without any of these three processes requiring the others to halt.

Fifth: The frame operates at every structural level per B1.20 recursive inheritance. Mutation, directed selection, and action-feedback all apply at the cell level, the aspect level, and the Self level. The three-mechanism architecture is not a cell-only specification. Different levels can evolve at different rates, governed by humans with different authority scopes, but the same three-mechanism architecture applies at each level. The integrating frame is recursive — what holds for a cell holds for an aspect composed of cells and for a Self composed of aspects.

Sixth: All DNA changes produced by any of the three mechanisms follow A6.02 rule retroactivity treatment. When directed selection produces a DNA layer change, A6.02 governs how the change applies to existing substrate content. When action-feedback evolution produces a DNA layer change, A6.02 governs the same way. Even when mutation integration leads to capability supersession and a resulting DNA layer restructuring, A6.02 governs the restructuring. The integrating frame holds retroactivity as a uniform commitment across all three evolution pathways that can produce DNA changes.

3. What makes the three mechanisms integrating frame architecturally distinctive

The most important contrast is with conventional AI system architectures. The dominant evolution model for AI systems is single-mechanism: model retraining. The system accumulates feedback, the model is retrained on updated data, and the system “evolves” in the sense that the model’s parameters

shift. This is one mechanism operating on one layer. There is no distinction between undirected variation, directed improvement, and evidence-informed refinement — all three are collapsed into a single retraining cycle, or some subset is addressed while the others go unarchitected.

CKS has three explicitly architected evolution mechanisms covering three distinct evolutionary dimensions. The instinct/reasoning separation (Paper 2 Claim 1) is what makes this possible: because the instinct layer and the DNA layer are architecturally distinct, they can evolve through distinct mechanisms under distinct governance shapes. A single-layer AI system has nowhere to locate this distinction; it has one evolvable object and one evolution mechanism by default. The CKS three-mechanism architecture is a direct architectural consequence of the instinct/reasoning separation — the separation is not merely a runtime property but an evolutionary property.

The integrating frame is what makes the multi-mechanism architecture coherent rather than merely additive. It would be possible, in principle, to specify three evolution mechanisms for a CKS deployment without specifying how they relate, how they operate together, or what makes them one architecture. The result would be three processes that happen to coexist. The integrating frame — specifically the concurrent-operation specification and the distinct-dimensions specification — is what transforms three coexisting processes into one multi-dimensional evolution architecture. This is the architectural commitment that the integrating frame formalizes and that has no exact parallel in published AI system architectures.

4. The biological analog and where CKS exceeds it

The three mechanisms have loose parallel in biological evolutionary processes. Mutation in the CKS sense parallels random genetic mutation — undirected variation arriving from processes the organism does not control. Directed selection parallels selective breeding or human-directed evolution — deliberate improvement toward goals with explicit selection criteria. Action-feedback evolution has a loose parallel in epigenetic inheritance from environmental experience — accumulated operational experience influencing heritable substrate content, though CKS’s action-feedback mechanism is explicitly governed rather than epigenetically automatic.

The biological parallel is useful as a conceptual scaffold but should not be pressed further than it supports. Biological evolution does not have an integrating frame for its evolutionary processes. Natural selection, genetic drift, epigenetic inheritance, niche construction, and the other mechanisms recognized by the Extended Evolutionary Synthesis operate without an explicit architectural specification relating them. They compose empirically over geological time through the dynamics of fitness landscapes and population genetics, not through a specified integrating architecture. CKS exceeds biology on this point: the three mechanisms are explicitly architected as one evolution architecture from the deployment’s inception. A CKS deployment does not need geological time to discover that its three evolution mechanisms compose — the integrating frame specifies the composition at design time.

CKS also exceeds biology in having directed selection as an explicit mechanism at all. Biology cannot have directed selection because biology has no governance-defined goals; natural selection operates without direction toward pre-specified objectives. CKS has directed selection because CKS has human

governance, and governance-defined goals are themselves substrate content that humans can modify. The presence of directed selection as an explicit mechanism alongside mutation is the first point at which the biological parallel breaks down as more than metaphor.

5. Inherited Paper 1 and Series A commitments

The three mechanisms integrating frame inherits several foundational commitments that apply across all three mechanisms without exception.

A1.01 (human-governed) applies across all three mechanisms, though the governance shape differs per mechanism. Human governance of mutation operates through verification, routing, and pinning. Human governance of directed selection operates through the standard authority architecture. Human governance of action-feedback operates through two-stage human mediation. In all three cases, the authority-vs-labor distinction from Paper 1 holds: humans hold authority over what evolves, in what direction, and through what processes; the labor of authoring, proposing, or integrating is allocable across humans and LLMs operating under human direction.

A6.02 (rule retroactivity) applies to all DNA changes produced through any evolution pathway, as stated in the sixth property of the integrating frame. Retroactivity treatment is not mechanism-specific; it follows from any change that modifies the DNA layer.

A2.40 (provenance) applies to all evolution events. Every mutation integration, every directed selection change, every action-feedback proposal and approval is recorded with provenance. The integrating frame does not create exceptions to provenance for any mechanism.

A1.07 (retraceability) applies through the entire evolution history. A CKS deployment's evolution history — across all three mechanisms and all structural levels — is preserved and retraceable. The integrating frame specifies that evolution events from all three mechanisms compose into one retraceable history, not three separate histories.

B1.20 (recursive inheritance) applies as the fifth property of the integrating frame. The three-mechanism architecture applies at every structural level. Cell-level evolution, aspect-level evolution, and Self-level evolution all operate through the same three mechanisms at appropriate scope.

6. Operational implications

A CKS deployment operating under the integrating frame has several operationally significant properties.

The deployment operates all three mechanisms concurrently as a standing architectural property. This is not a special mode; it is the normal operating condition. At any point in the deployment's operational life, an LLM upgrade may arrive requiring mutation integration under verification; a human governance decision may produce a DNA layer change under directed selection; accumulated action evidence may surface proposals for DNA refinement under action-feedback. The integrating frame specifies that the deployment handles all three without requiring any of them to wait for the others.

The three mechanisms are not equally active at all times. Mutation is tied to the LLM vendor's upgrade cadence — it occurs when upstream upgrades arrive, not on a schedule the deployment controls. Directed selection follows human governance deliberation — it occurs when humans with appropriate authority decide to modify the DNA layer. Action-feedback accumulates based on operational throughput — its proposals become substantive only after enough action evidence has accumulated to ground meaningful proposals. A deployment's evolutionary activity at any given time reflects which mechanisms happen to be active in that period. The integrating frame is what allows deployment operators to reason coherently about which evolutionary dimension is most active and what governance processes are currently running.

Different cell types within a deployment may rely on different mechanisms more heavily. A cell type with a stable, well-understood function may evolve primarily through mutation as the LLM layer sharpens its underlying capability. A cell type in a rapidly changing domain may evolve primarily through directed selection as humans actively update its orchestration rules. A cell type with high operational throughput may evolve primarily through action-feedback as operational patterns accumulate into substantive proposals. The integrating frame enables deployment designers to reason about these differences without treating each cell type's evolution as an isolated phenomenon.

7. Limits of the integrating frame

The three mechanisms integrating frame does not prescribe a specific mechanism activation schedule. The cadence of mutation integration, directed selection, and action-feedback proposals is a deployment-specific operational matter, not a specification the integrating frame makes.

The frame does not collapse the three mechanisms into one. Each mechanism retains its distinct character — mutation remains undirected variation on the instinct layer, directed selection remains directed improvement on the DNA layer, action-feedback remains evidence-informed refinement on the DNA layer through action evidence. The frame specifies how they compose; it does not homogenize them.

The frame is not itself an evolution mechanism. It does not produce changes in the CKS architecture. It specifies the architectural relationships among three mechanisms that do produce changes. Confusing the frame with one of the mechanisms it relates would be a category error.

The frame does not eliminate per-mechanism governance shapes. The mechanism-appropriate governance shapes (verification for mutation, authority architecture for directed selection, two-stage human mediation for action-feedback) are specified per mechanism. The integrating frame specifies that all three are governed; the per-mechanism shapes are the subject of B2.59 (cross-mechanism governance), which develops the governance machinery across the three mechanisms in detail.

The frame applies to CKS deployments. It is not a general evolutionary framework claim for AI systems at large. CKS's three-mechanism architecture follows from the instinct/reasoning separation and the DNA/action layer distinction; these are CKS-specific architectural commitments. The integrating frame is the specification of how those commitments compose into a multi-dimensional evolution architecture within the CKS pattern.

The three mechanisms are not claimed to be exhaustive of all possible evolution-like processes in a CKS deployment. A deployment may have additional processes that change system behavior over time. What the integrating frame specifies is the three architecturally named and governed evolution mechanisms per B1.12; other processes are not within the frame's scope.

8. One-sentence test

A CKS deployment instantiates the three mechanisms integrating frame if and only if: mutation (instinct evolution), directed selection (DNA evolution), and action-feedback evolution are each specified as operating on their respective architectural layers under mechanism-appropriate governance shapes, all three can operate concurrently without mutual interference, all three apply at every structural level, and all DNA changes produced through any of the three mechanisms are subject to A6.02 retroactivity treatment.

9. Why naming as standalone matters; opening the B1.12 decomposition

The three mechanisms integrating frame is nameable as a standalone architectural commitment because it is the object that distinguishes a multi-mechanism evolution architecture from three coexisting evolution processes. Without the integrating frame, a CKS deployment might have all three mechanisms and still lack the architectural specification that makes them one architecture. The frame is the difference between a deployment that happens to have three evolution processes and a deployment that has a three-dimensional evolution architecture.

Naming the integrating frame as a standalone derivation note serves the defensive publication purpose of the Series B chain. The individual mechanism notes (B1.13, B1.14, B1.15 decompositions) will formalize mutation, directed selection, and action-feedback as separate architectural commitments. B2.56 formalizes the integrating frame — the architectural object that relates them — as prior art separately from any of the three. Any claim to novel invention in multi-mechanism AI evolution architecture will need to account for the integrating frame as prior art, not merely for the individual mechanism specifications.

This note is the fifty-sixth Phase B2 note and the first of the five-note B1.12 decomposition. B2.57 will formalize productive tension as an operational specification — what it means for the three mechanisms to compose productively rather than destructively. B2.58 will formalize mechanism priority and sequencing — how deployments reason about which mechanism takes precedence when two mechanisms produce potentially conflicting evolutionary pressures. B2.59 will formalize cross-mechanism governance — the full specification of how the three distinct governance shapes compose under the integrating frame. B2.60 will formalize three mechanisms verification — the operational tests that confirm a deployment has correctly instantiated the integrating frame. After B2.60, Phase B2 continues with the B1.13 mutation decomposition at B2.61–B2.66 and the decompositions that follow.

Derivation note B2.56 in the CKS Series B chain. Derives from Paper 2 §7 (Claim 4) and §8 (Claim 5). Directly load-bearing references: B1.12 (three evolution mechanisms in productive tension), B1.13 (mutation), B1.14 (directed selection), B1.15 (action-feedback evolution), A1.01 (human-governed), A6.02 (rule retroactivity),

A2.40 (provenance), A1.07 (retraceability), B1.20 (recursive inheritance). Forthcoming: B2.57 (productive tension), B2.58 (mechanism priority and sequencing), B2.59 (cross-mechanism governance), B2.60 (three mechanisms verification).